

SmartFarm AI: An Integrated Season-Aware Decision Support and Financial Analytics Platform for Smallholder Agriculture

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Abstract

Digital agriculture research has established strong evidence for data-driven decision support, machine learning, and precision farming workflows; however, many practical deployments remain fragmented across advisory, field records, and farm economics [1]-[7]. This paper presents **SmartFarm AI**, a full-stack platform designed to unify farm profile management, season-aware financial records, crop-level profitability comparison, and corrective edit workflows in a single operational interface. The implemented system uses a lightweight architecture (React + TypeScript frontend, Node.js + Express services, SQLite database), enabling low-infrastructure deployment while preserving analytical depth. A period-aware schema was implemented for expenses and revenues using explicit date-season-year fields, followed by monthly and season-wise aggregation, same-crop inter-season comparison, and rule-based recommendation generation. Prototype validation shows improved interpretability of farm economics by separating overall totals from scope-filtered analytics and by enabling post-entry correction without losing data consistency.

Index Terms

Smart farming, precision agriculture, farm decision support, seasonal profitability analytics, machine learning in agriculture, digital agriculture systems.

I. Introduction

Agriculture is increasingly influenced by climate variability, input-price volatility, and market uncertainty. Contemporary smart-farming research demonstrates that data-intensive technologies can improve planning quality and operational outcomes [1], [8]. In parallel, deep learning and machine learning have broadened the predictive scope of agricultural systems, including crop health analysis, yield support, and optimization tasks [2], [3], [6], [7].

Despite these advances, real-world farm software often remains functionally fragmented. Operational records, crop advisories, and financial performance are frequently maintained in disconnected tools, limiting actionable interpretation at season boundaries. The practical question for farmers is not only what to plant, but also which crop remained profitable after full seasonal costs and how current margin compares with the previous season for the same crop.

This work addresses that operational gap through SmartFarm AI, with emphasis on period-aware financial intelligence and correction-safe data workflows.

II. Related Work and Gap Analysis

A. Literature Positioning

Big-data-centric smart farming architectures have been reviewed extensively, with strong emphasis on integration, analytics, and ecosystem interoperability [1]. Survey evidence also confirms the growing role of deep learning and machine learning in agricultural decision pipelines [2], [3], [6], [7]. IoT and sensing layers further support precision interventions and contextual monitoring [4], [5].

B. Gap Statement

From a deployment perspective, three practical gaps persist:

1. Limited season-year linkage in transactional farm records.
2. Weak same-crop period-over-period profitability comparison.
3. Inadequate post-entry correction mechanisms in financial modules.

In addition, trust-centered concerns (data governance, clarity, and user confidence) remain central to digital agriculture adoption [9].

C. Contribution Summary

1. A unified period-aware financial model for expense and revenue tracking.
2. Crop-specific inter-season comparison (same crop, consecutive periods).
3. A correction-enabled edit workflow with immediate analytics recomputation.
4. A lightweight architecture suitable for low-resource deployment.

III. Materials and Methods

A. Research Design

A design-and-implementation methodology was used: requirement extraction from practical dashboard limitations, end-to-end data model extension with season/year metadata, API and UI integration for create-read-update financial flows, and analytical validation through controlled usage scenarios.

B. System Stack

- Frontend: React, TypeScript, Vite.
- Backend: Node.js, Express REST APIs.
- Persistence: SQLite.
- Security: JWT authentication and bcrypt password hashing.

C. Data and Computation Model

For each transaction, the period context is stored as `date`, `season`, and `season_year`. Core equations are defined as: $NetProfit = TotalRevenue - TotalExpenses$; $PeriodDelta = Profit(CurrentSeason) - Profit(PreviousSeason)$; and $CategoryShare = CategoryExpense / TotalExpense$.

IV. Implementation and Analytical Workflow

The implementation includes schema and route upgrades for expenses and payments entities, plus frontend controls for date/season/year capture. The dashboard explicitly separates overall totals from scope-filtered totals to reduce metric ambiguity during interpretation.

Module	Issue in Legacy Workflows	Implemented Improvement
Financial schema	No season-year normalization	Added <code>date</code> , <code>season</code> , <code>season_year</code> fields
Comparison logic	No same-crop prior-season comparison	Current vs previous season crop delta
Data quality	Add-only records	Edit flow with recomputation support
Interpretation	Scope-vs-overall mismatch confusion	Explicitly separated overall and scoped totals

V. Results and Discussion

Observed prototype outcomes include improved visibility of seasonal profitability, direct same-crop cross-season comparison, safer correction workflow behavior, and stronger cash-flow interpretation from received versus pending sales records. These observations are consistent with prior literature that emphasizes deployment realism and explainability [1], [4], [7], [9].

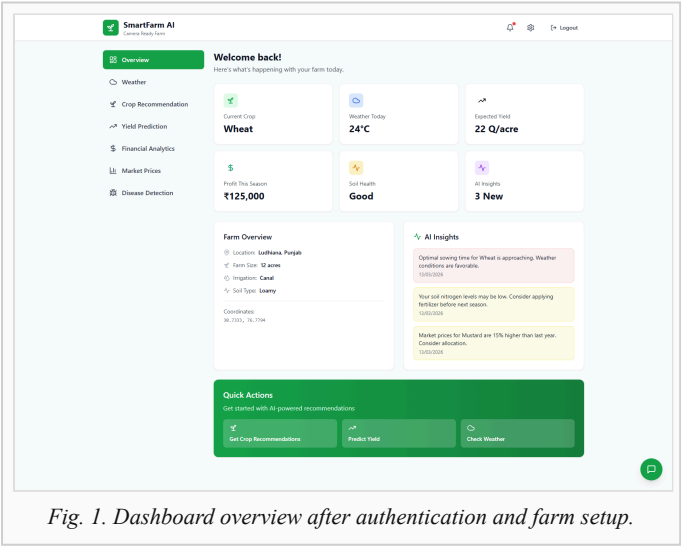


Fig. 1. Dashboard overview after authentication and farm setup.

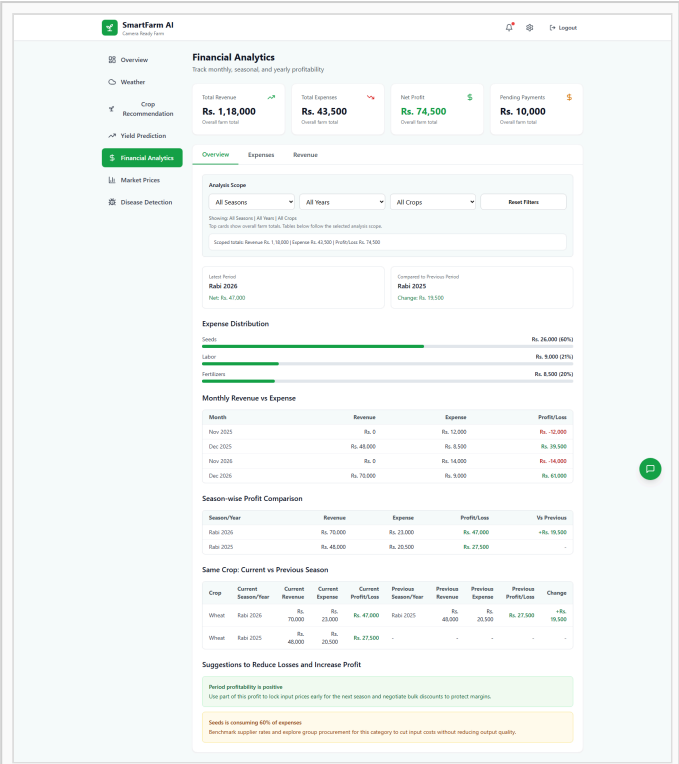


Fig. 2. Financial analytics with overall KPI cards and scoped analysis filters.

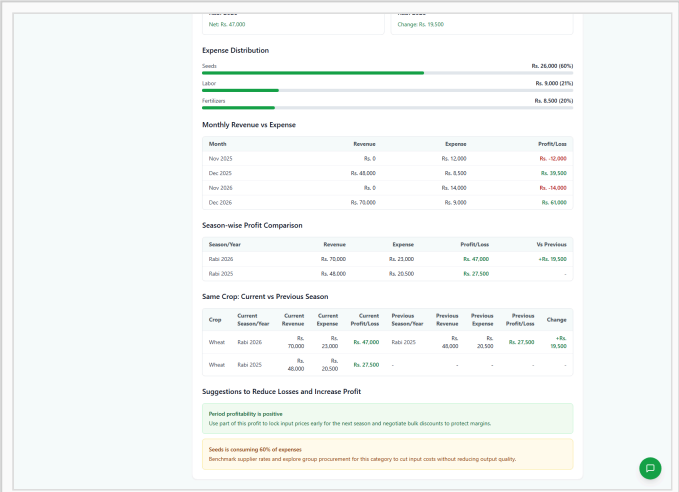


Fig. 3. Same-crop current versus previous season profitability comparison.

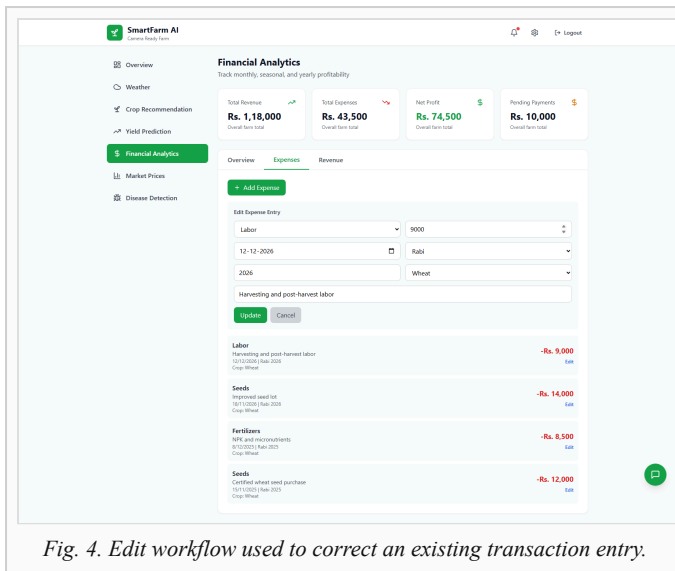


Fig. 4. Edit workflow used to correct an existing transaction entry.

VI. Conclusion

This study delivers an integrated and deployable smart-farming implementation that couples advisory-ready platform structure with season-aware financial analytics. By introducing explicit period metadata, same-crop inter-season comparison, and correction-enabled workflows, SmartFarm AI closes a practical gap between agricultural AI concepts and farmer-facing decision support.

Future work will extend from rule-based recommendations to model-driven forecasting and will integrate live weather, market, and supply-chain intelligence in a unified decision loop [10].

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All figures are generated from the running SmartFarm AI prototype and captured automatically in this workspace.